

A RECAP OF THE MISSING VALUE ANALYSIS Evaluation of the issues surrounding missing data in the data set can be summarized in four conclusions:

1. **The missing data process is MCAR.** All of the diagnostic techniques support the conclusion that no systematic missing data process exists, making the missing data MCAR (missing completely at random). Such a finding provides two advantages to the researcher. First, it should not involve any hidden impact on the results that need to be considered when interpreting the results. Second, any of the imputation methods can be applied as remedies for the missing data. Their selection need not be based on their ability to handle nonrandom processes, but instead on the applicability of the process and its impact on the results.
2. **Imputation is the most logical course of action.** Even given the benefit of deleting cases and variables, the researcher is precluded from the simple solution of using the complete case method, because it results in an inadequate sample size. Some form of imputation is therefore needed to maintain an adequate sample size for any multivariate analysis.
3. **Imputed correlations differ across techniques.** When estimating correlations among the variables in the presence of missing data, the researcher can choose from four commonly employed techniques: the complete case method, the all-available information method, the mean substitution method, and the EM method. The researcher is faced in this situation, however, with differences in the results among these methods. The all-available information, mean substitution, and EM approaches lead to generally consistent results. Notable differences, however, are found between these approaches and the complete information approach. Even though the complete information approach would seem the most “safe” and conservative, in this case it is not recommended due to the small sample used (only 26 observations) and its marked differences from the other two methods. The researcher should, if necessary, choose among the other approaches.
4. **Multiple methods for replacing the missing data are available and appropriate.** As already mentioned, mean substitution is one acceptable means of generating replacement values for the missing data. The researcher also has available the regression and EM imputation methods, each of which give reasonably consistent estimates for most variables. The presence of several acceptable methods also enables the researcher to combine the estimates into a single composite, hopefully mitigating any effects strictly due to one of the methods.

In conclusion, the analytical tools and the diagnostic processes presented in the earlier section provide an adequate basis for understanding and accommodating the missing data found in the pretest data. As this example demonstrates, the researcher need not fear that missing data will always preclude a multivariate analysis or always limit the generalizability of the results. Instead, the possibly hidden impact of missing data can be identified and actions taken to minimize the effect of missing data on the analyses performed.

OUTLIERS

Outliers are observations with a *unique combination of characteristics identifiable as distinctly different* from the other observations. What constitutes a unique characteristic? Typically it is judged to be an unusually high or low value on a variable or a unique combination of values across several variables that make the observation stand out from the others. In assessing the impact of outliers, we must consider the practical and substantive considerations:

- From a *practical* standpoint, outliers can have a marked effect on any type of empirical analysis. For example, assume that we sample 20 individuals to determine the average household income. In our sample we gather responses that range between \$20,000 and \$100,000, so that the average is \$45,000. But assume that the 21st person has an income of \$1 million. If we include this value in the analysis, the average income increases to more than \$90,000. Obviously, the outlier is a valid case, but what is the better estimate of the average household

income: \$45,000 or \$90,000? The researcher must assess whether the outlying value is retained or eliminated due to its undue influence on the results.

- In *substantive* terms, the outlier must be viewed in light of how representative it is of the population. Again, using our example of household income, how representative of the more wealthy segment is the millionaire? If the researcher feels that it is a small, but viable segment in the population, then perhaps the value should be retained. If, however, this millionaire is the only one in the entire population and truly far above everyone else (i.e., unique) and represents an extreme value, then it may be deleted.

Outliers cannot be categorically characterized as either beneficial or problematic, but instead must be viewed within the context of the analysis and should be evaluated by the types of information they may provide. When beneficial, outliers—although different from the majority of the sample—may be indicative of characteristics of the population that would not be discovered in the normal course of analysis. In contrast, problematic outliers are not representative of the population, are counter to the objectives of the analysis, and can seriously distort statistical tests. Owing to the varying impact of outliers, it is imperative that the researcher examine the data for the presence of outliers and ascertain their type of influence. The reader is also referred to the discussions in Chapter 4 that relate to the topic of influential observations. In these discussions, outliers are placed in a framework particularly suited for assessing the influence of individual observations and determining whether this influence is helpful or harmful.

Why do outliers occur? Outliers can be classified into one of four classes based on the source of their uniqueness:

- The first class arises from a *procedural error*, such as a data entry error or a mistake in coding. These outliers should be identified in the data cleaning stage, but if overlooked they should be eliminated or recorded as missing values.
- The second class of outlier is the observation that occurs as the result of an *extraordinary event*, which accounts for the uniqueness of the observation. For example, assume we are tracking average daily rainfall, when we have a hurricane that lasts for several days and records extremely high rainfall levels. These rainfall levels are not comparable to anything else recorded in the normal weather patterns. If included, they will markedly change the pattern of the results. The researcher must decide whether the extraordinary event fits the objectives of the research. If so, the outlier should be retained in the analysis. If not, it should be deleted.
- The third class of outlier comprises *extraordinary observations* for which the researcher has no explanation. In these instances, a unique and markedly different profile emerges. Although these outliers are the most likely to be omitted, they may be retained if the researcher feels they represent a valid element of the population. Perhaps they represent an emerging element, or an untapped element previously not identified. Here the researcher must use judgment in the retention/deletion decision.
- The fourth and final class of outlier contains observations that fall within the ordinary range of values on each of the variables. These observations are not particularly high or low on the variables, but are *unique in their combination* of values across the variables. In these situations, the researcher should retain the observation unless specific evidence is available that discounts the outlier as a valid member of the population.

Detecting and Handling Outliers

The following sections detail the methods used in detecting outliers in univariate, bivariate, and multivariate situations. Once identified, they may be profiled to aid in placing them into one of the four classes just described. Finally, the researcher must decide on the retention or exclusion of each outlier, judging not only from the characteristics of the outlier but also from the objectives of the analysis.